APL LOGISTICS

Customer, Product & Profitability Performance Analysis

Business Analyst Internship Project Report

Project focus
Customer value • Product profitability • Discount-driven margin erosion • Actionable business insights

Prepared from the uploaded APL_Logistics.csv dataset

Prepared for internship reporting and business analysis discussion

1. Executive Summary

This project examines how transaction-level analytics can support commercial decision-making by connecting customer value, product profitability, discounting, and data quality. Rather than focusing only on operational speed or efficiency, the analysis emphasizes how profitably sales and customer relationships perform.

The preliminary results indicate that profitability is concentrated among a subset of customers, some product lines have negative recorded profit, and average recorded profit differs across discount bands. These patterns can help prioritize further investigation, but they should not be treated as proof of causation or as verified financial results of APL Logistics.

Important data context: the uploaded file contains retail-style product transactions. The available file alone does not establish that these are proprietary APL Logistics operating or financial records. Figures below refer to values recorded in the CSV.

2. Dataset Overview

Metric	Reported value	Interpretation
Transaction line items	180,519	Rows/line items in the dataset
Unique customers	20,652	Distinct customer IDs
Recorded sales	$36.78 million	Sum of the sales field
Recorded profit	$3.97 million	Sum of the profit field
Recorded discounts	$3.73 million	Sum of the discount amount field
Profit-to-sales ratio	10.78%	Recorded profit divided by recorded sales

These are preliminary reported values and should be reconciled against the dataset's field definitions and business rules before formal use.

3. Customer Value Analysis

Customer value analysis identifies customers who contribute strongly to recorded profit and customers whose cumulative transactions show losses. The dataset contains 20,652 distinct customers. Based on the preliminary analysis, the top 20% of customers ranked by recorded profit contribute approximately 79.2% of total recorded profit and 40.8% of total sales. Approximately 4,069 customers have negative cumulative recorded profit.

Business interpretation

The pattern suggests that customer profitability may be concentrated among a relatively small group. Management could use customer segmentation to support retention, account planning, and review of loss-making relationships. Negative cumulative profit should trigger investigation rather than an automatic decision to discontinue a customer.

Recommended actions

• Prioritize retention and relationship management for consistently profitable customers.
• Investigate customers with negative cumulative profit, considering discounts, product mix, service costs, returns, and possible data errors.
• Assess customer profit alongside sales, purchase frequency, service requirements, and long-term potential.

Suggested KPI: Customer Profit Contribution = customer recorded profit ÷ total recorded profit × 100. Interpret carefully when total profit is small or negative.

4. Product and Category Profitability

The category-level results identify the largest contributors to total recorded profit. These rankings reflect total profit, not necessarily the highest profit margin.

Category	Recorded profit	Share of total profit (approx.)
Fishing	$756,221	19.1%

Category	Recorded profit	Share of total profit (approx.)
Cleats	$494,637	12.5%
Camping & Hiking	$427,456	10.8%
Cardio Equipment	$383,011	9.7%
Women's Apparel	$350,421	8.8%

Products with negative cumulative recorded profit

Product	Recorded profit
SOLE E35 Elliptical	–$965.12
Bushnell Pro X7 Jolt Slope Rangefinder	–$255.95
SOLE E25 Elliptical	–$169.56

Recommended actions: prioritize profitable categories where demand and inventory conditions support it; review low-margin categories; and audit the costs, pricing, returns, and transactions of loss-making products before making assortment decisions.

Total profit and profit margin are different measures. A high-volume category can generate high total profit while still having a comparatively low margin.

5. Discount-Driven Margin Erosion

The dataset records approximately $3.73 million in discounts, equivalent to about 10.14% of recorded sales. The average recorded discount rate is approximately 10.17%. Average recorded profit per transaction line differs across discount levels.

Discount rate	Average profit per line	Average profit ratio
0%	$26.67	12.75%
5%	$23.78	12.65%
10%	$22.13	11.94%
15%	$20.06	11.63%
20%	$17.68	11.23%
25%	$19.13	12.72%

Comparing the 0% and 20% discount groups, average recorded profit per line is approximately 33.7% lower in the 20% group. This is an association, not proof that discounts alone caused the difference. Product mix, customer mix, and other factors may explain some of the variation.

Recommended discount controls

- Define minimum margin thresholds by product or category.
- Require approval for discounts above agreed limits.
- Compare discounted and non-discounted transactions for similar products and customer groups.
- Measure incremental profit from promotions, not only sales uplift.
- Include delivery, handling, returns, and other relevant costs when these data are available.

6. Data Quality and Reliability Checks

The preliminary review identifies several issues that should be resolved before the results are used for formal financial decisions:

• 33,784 transaction lines (approximately 18.7%) have negative recorded profit.
• Some records contain losses much larger than sales value; for example, one record shows sales of $1,999.99 and profit of –$4,274.98.
• Canceled, suspected-fraud, and other non-final order statuses also have recorded profit totals.
• Product names and category assignments may require validation.

Recommended validation steps

1. Confirm the definition and calculation of the profit field.
2. Check duplicate records, missing values, extreme values, and product mappings.
3. Confirm how canceled, returned, and suspected-fraud transactions should be treated.
4. Recalculate metrics using an agreed set of valid transactions.
5. Document all exclusions and assumptions in the final report.

7. Additional Commercial and Operational Insights

The recorded sales total is approximately $36.78 million and recorded profit is approximately $3.97 million, producing a profit-to-sales ratio of about 10.78%. This is a preliminary ratio, not necessarily an accounting margin, until field definitions and cost treatment are confirmed.

The dataset also includes shipping and late-delivery indicators. A follow-up analysis could test whether delivery performance is associated with customer profitability or product returns. Any such relationship should be tested rather than assumed to be causal.

The available file does not clearly support reliable time-trend analysis or order-level analysis unless appropriate order identifiers and dates are present and validated.

8. Recommended Business Action Plan

Priority	Action	Expected benefit
1	Validate profit values, order status, and product mappings	More reliable reporting
2	Segment customers by cumulative profit and sales	Better retention and account decisions
3	Review category and product margin	Improved portfolio and pricing decisions
4	Introduce margin-based discount thresholds	Reduced avoidable margin erosion
5	Compare delivery metrics with profit outcomes	Joined-up commercial and operational view

9. Recommended Dashboard KPIs

KPI	Purpose
Total sales	Measure recorded revenue
Total recorded profit	Monitor reported profitability
Profit-to-sales ratio	Compare profitability relative to sales
Customer profit contribution	Identify high-contribution customers
Product/category profit	Compare product performance
Average discount rate	Monitor discount intensity
Negative-profit transaction rate	Flag potential losses and data errors
Profit by discount band	Investigate discount-profit relationships
Late-delivery rate	Monitor service performance

10. Conclusion

The analysis of the APL_Logistics.csv dataset demonstrates how customer, product, and discount-level analytics can support more informed commercial decision-making. The preliminary results indicate that recorded profitability is concentrated among a smaller group of customers, product performance varies across categories, and discount bands show differences in average recorded profit. These findings highlight opportunities for customer segmentation, portfolio review, and margin-focused pricing strategies.

The analysis also identifies data-quality concerns, including negative-profit transactions and unusual financial values, that require validation before the results can support formal decisions. By combining customer value analysis, product profitability assessment, discount monitoring, and data validation, the project illustrates how analytics can connect commercial performance with sustainable business growth.

Report note: All values in this document are based on preliminary analysis figures and should be independently reconciled with the raw CSV and the dataset's field definitions before submission as audited or official business results.